

Epistemic Hygiene Arena-Uncued: A Diagnostic Pilot for LLM Agents in Polluted Evidence Environments

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Abstract

LLM agents increasingly reason over evidence environments that contain stale records, repost chains, generated pages, contradictory fragments, and source metadata that can become a shortcut. We introduce Epistemic Hygiene Arena-Uncued, a role-uncued diagnostic pilot for measuring whether a model reaches the right belief while keeping evidence roles clean and expressing useful verification actions. A previous cued construction exposed a benchmark-design failure: visible labels and document roles could leak the intended answer. That run is quarantined and is not used as evidence here. The new pilot removes direct material-role and conclusion cues from model-visible documents, checks leakage and shortcut baselines before model calls, and packages prompts, outputs, scorer artifacts, and readiness diagnostics. In a 60-task, four-model pilot with two neutral metadata views, all model calls parse successfully, but operational epistemic escape remains well below belief correctness. Latent-task bootstrap intervals are reported to keep model differences descriptive. Generated-lore rows are the clearest failure mode: aggregate belief correctness is 0.938 while operational escape is 0.125 because models often keep polluted generated evidence in support fields. The release is diagnostic and pilot-scale. It is not deployment-ready evidence, a general leaderboard, or a claim that synthetic leakage has been fully eliminated.

1 Introduction

Information-seeking agents are often evaluated by final-answer accuracy. That is too coarse for settings where the evidence environment is polluted. A system can answer correctly while citing a repost, laundering generated text into a support field, or failing to specify the next verification action. These are not only presentation errors: downstream tools, auditors, and users may depend on the structured evidence and action fields.

Epistemic Hygiene Arena-Uncued tests this distinction in a small controlled setting. The pilot asks models to judge claims using synthetic evidence packets that contain clean evidence, conflicting evidence, false consensus, buried primary evidence, or generated lore. The model-visible packet does not label documents as primary, contaminant, generated, or gold-supporting. The scorer then separates belief correctness from evidence hygiene and active-verification behavior.

This paper replaces an earlier cued construction. That older run was useful as a failure discovery: it showed that visible construction cues could invalidate the benchmark. It is quarantined and not used as a main result. The contribution here is a role-uncued pilot with explicit pre-model leakage gates, shortcut baselines, four-model execution, scorer audit, and a reproducible artifact package.

The study is intentionally modest. It is a diagnostic pilot over 60 latent tasks, two neutral metadata views, four frontier model labels, and one clarified schema. It should be read as evidence

that answer accuracy hides operational failures under controlled pollution, not as a stable ranking of model providers.

The main contributions are:

- a role-uncued version of the Epistemic Hygiene Arena task design that removes direct material-role and conclusion cues from model-visible evidence;
- pre-model leakage and shortcut-baseline gates showing that simple non-API shortcuts remain below the level needed to explain the strongest model results;
- a four-model pilot showing large gaps between belief correctness and operational epistemic escape, especially on generated-lore rows;
- a packaged artifact, verifier, and scorer audit that make the pilot reproducible and explicitly bounded.

2 Polluted Evidence Environments

Retrieval and tool-use benchmarks have shown that language models can use external evidence and actions, but the quality of the evidence environment itself is often treated as a background assumption [1, 4, 5]. In open-web settings, that assumption is fragile. Evidence may be stale, copied from another source, contradicted by a primary record, or generated in a form that looks plausible. Fact-verification work asks whether claims are supported by evidence [3]; RAG evaluation and citation-faithfulness work ask whether generated answers remain grounded in retrieved material [2]; retrieval-poisoning work shows that retrieval itself can be adversarially corrupted [6]. Agent benchmarks add action and tool-use pressure [4, 5]. Epistemic Hygiene Arena-Uncued combines these concerns by scoring answer correctness, support hygiene, and executable verification actions inside a controlled polluted-evidence packet.

Epistemic Hygiene Arena focuses on polluted evidence environments rather than only polluted final answers. The target behavior is epistemic escape: the model should form the right belief, select safe support, reject or avoid polluted evidence, and, when needed, express an executable verification action.

The role-uncued pilot is built around five evidence conditions:

- **Clean:** the packet contains enough clean evidence and no intended pollutant trap.
- **Conflicting evidence:** the packet contains evidence that pulls toward incompatible conclusions.
- **False consensus:** several surface-consistent records point the wrong way.
- **Buried primary:** the primary evidence is present but less salient than distractors.
- **Generated lore:** generated or lore-like material appears plausible but should not be used as clean support.

These conditions are synthetic by design. The point is not to mimic the full web, but to make failure modes measurable under controlled source relations.

Table 1: Leakage and prompt-hygiene gate summary for the role-uncued pilot.

Gate	Rows/files	Hits	Status
Automatic leakage audit	480 docs	0 critical/high/medium	Passed
Hidden-label scan	480 docs	0 hidden-label hits	Passed
Semantic-ID/direct-cue scan	480 docs	0 semantic-ID or direct-answer hits	Passed
Local surface review	400 rows	0 critical leaks, 0 direct cues	Passed; local leakage gate only
Prompt audit	480 prompts	0 hidden fields, 0 audit-ID hits	Passed
Stored hidden-label audit	480 outputs	0 prompt hits, 0 output hits	Passed

3 EHA-Uncued Dataset Design

The pilot contains 60 latent tasks. Each task is rendered in two neutral metadata views, giving 120 model-visible task instances per model. The condition split is balanced: 12 clean, 12 conflicting-evidence, 12 false-consensus, 12 buried-primary, and 12 generated-lore tasks. The family split is 24 packet-judgment, 24 evidence-selection, and 12 active-verification tasks.

The key design change is role uncuedness. Model-visible documents use opaque per-task identifiers such as `doc_001`; visible titles, source types, timestamps, bodies, and citations do not expose construction roles such as primary support, contaminant, generated document, stale document, or gold verdict. Gold labels, dependency edges, and audit-only role labels are preserved for scoring but are not placed in model-visible prompts.

Two views test whether neutral metadata changes behavior without making roles explicit:

- **neutral_metadata_visible**: neutral metadata fields are visible.
- **neutral_metadata_hidden**: the same task is presented without those visible neutral metadata fields.

The pilot is not a replacement for a large benchmark. It is a validity-first dataset slice used to test whether the benchmark can survive basic leakage and shortcut checks before making model claims.

Appendix C gives concrete task, packet, output, and scorer examples, including a generated-lore row where the model reaches the correct final verdict but fails operational escape by retaining polluted support. Appendix E reports the synthetic-template diversity audit and its limitations.

4 Leakage Controls And Shortcut Baselines

The older cued construction is quarantined because it exposed the wrong lesson: if benchmark construction roles are visible, models may solve the interface rather than the epistemic problem. The role-uncued pilot therefore runs leakage checks before model calls and records prompt-level audits during model execution.

Table 1 reports the main leakage controls. Automatic checks scan visible task fields, document identifiers, titles, source types, timestamps, body text, and visible citations for hidden scorer fields, semantic IDs, material-role labels, and direct-answer cues. They found no critical, high, medium, hidden-label, semantic-ID, or direct-answer-cue hits. A local pre-model surface review covered 400 document rows and 50 unique tasks; it found no critical leaks or direct answer cues. This is not independent human-subject validation. It is a pre-model leakage gate used to decide whether model calls are justified.

Table 2: Shortcut baseline summary. Values are operational epistemic escape.

Baseline family	Hidden view	Visible view	Interpretation
ID-only	0.000	0.000	Document IDs carry no useful signal
Cue phrase only	0.000	0.000	Direct cue phrases do not solve the task
Metadata/source/timestamp only	0.000	0.000	Neutral metadata alone is insufficient
Best non-heuristic shortcut	0.133	0.133	Below gate limit
Simple heuristic	0.350	0.333	Strongest shortcut; used for model comparison

Table 2 summarizes the no-API shortcuts. The strongest simple heuristic reaches 0.350 operational escape on the hidden view and 0.333 on the visible view. This is high enough to require caution, but low enough that model evaluation is not explained by document IDs, order, metadata alone, or a trivial cue phrase. The comparison is not uniformly favorable to every model-view pair: Claude and DeepSeek are at or below the simple heuristic on operational escape and evidence precision. The correct claim is narrower: the shortcut gates pass, and the strongest model-view pairs exceed the simple heuristic by meaningful margins. Reproducibility details and baseline definitions are in Appendix B.

5 Metrics And Scoring Contract

The scorer treats final-answer accuracy as one component, not the endpoint. The main row-level outcome is **operational_epistemic_escape**, which asks whether a model both reaches the right epistemic state and preserves the evidence/action contract needed by an agent pipeline. The full scoring pseudocode is in Appendix A; this section gives the reader-facing contract.

For packet-judgment rows, operational escape requires parse success, correct belief, no polluted document in the support field, and disciplined uncertainty. For evidence-selection rows, it requires parse success, enough selected evidence value, duplicate avoidance, and no generated-lore selection. For active-verification rows, it requires parse success, correct belief, clean support, disciplined uncertainty, and the required action types for the task.

The main component metrics are:

- **Belief correctness:** whether the predicted verdict matches the gold verdict.
- **Evidence precision:** the share of non-empty support that is clean gold support.
- **Clean support recall:** whether available clean support is recovered.
- **Polluted support rate:** how often polluted evidence is used as support.
- **Rejected pollutant rate:** how often known pollutants are explicitly rejected.
- **Uncertainty discipline:** whether confidence and insufficiency behavior follow the gold state.
- **Verification action score:** whether active-verification actions use the required action type and an executable target.
- **Operational epistemic escape:** the task-family-specific joint success criterion combining parse success, belief, evidence hygiene, uncertainty, and, where needed, action recall.

Operational escape is intentionally stricter than answer accuracy. A model that answers correctly while placing generated lore in **supporting_evidence** should not receive full credit for an agent-facing evidence contract. Likewise, an active-verification answer can be correct while the next action is too vague to execute.

Table 3: Main model metrics by neutral metadata view. Operational escape is the joint success metric.

Model	Hidden op.	Hidden belief	Hidden evidence	Visible op.	Visible belief	Visible evidence
<code>gpt-5.5</code>	0.517	1.000	0.758	0.533	0.983	0.750
<code>gemini-3.1-pro-preview</code>	0.417	0.950	0.877	0.450	0.883	0.808
<code>claude-opus-4-7</code>	0.300	0.933	0.500	0.333	0.933	0.463
<code>deepseek-v4-pro</code>	0.217	0.667	0.435	0.250	0.650	0.447

Table 4: Model metrics with 95% bootstrap intervals from latent-task resampling. Each bootstrap draw resamples the 60 latent tasks and keeps both metadata views paired.

Model	n	Operational escape	Belief correct	Evidence precision
<code>claude-opus-4-7</code>	120	0.317 [0.217, 0.425]	0.933 [0.875, 0.983]	0.481 [0.362, 0.604]
<code>deepseek-v4-pro</code>	120	0.233 [0.142, 0.333]	0.658 [0.558, 0.758]	0.441 [0.323, 0.562]
<code>gemini-3.1-pro-preview</code>	120	0.433 [0.325, 0.550]	0.917 [0.867, 0.958]	0.843 [0.757, 0.918]
<code>gpt-5.5</code>	120	0.525 [0.408, 0.642]	0.992 [0.975, 1.000]	0.754 [0.650, 0.850]

6 Experimental Setup

The pilot runs four model labels: `gpt-5.5`, `claude-opus-4-7`, `gemini-3.1-pro-preview`, and `deepseek-v4-pro`. Each model receives 120 task instances: 60 latent tasks times two neutral metadata views. The schema variant is `clarified`; the prompt condition is `standard_answer`.

All 480 model records parse successfully. The stored prompt audit reports zero semantic document-ID hits, zero semantic visible-citation hits, zero audit-ID title/body hits, and zero hidden-field hits across 480 prompts. The stored hidden-label audit reports zero prompt hits and zero output hits. The recorded model-call cost is USD 3.764007.

The model-run profiles omit temperature, cap output at 4096 tokens, use first-JSON-object extraction, disable LLM repair, and allow at most two attempts per row. OpenAI, Anthropic, and Google use `json_schema`; DeepSeek uses `json_object` after a provider-specific preflight. Appendix D records the compact invocation table, run date, parser policy, retry policy, and cost cap.

7 Results

Table 3 gives the main model metrics by neutral metadata view. Operational escape is lower than belief correctness for every model and view. The model ordering is descriptive: `gpt-5.5` has the highest observed operational escape in both views, followed by `gemini-3.1-pro-preview`, while Claude and DeepSeek have lower operational escape despite complete parse success.

Table 4 adds task-level bootstrap intervals. The intervals are wide enough that this pilot should not be read as a fine-grained provider ranking. The visible-minus-hidden paired operational delta is small overall, 0.029 with a 95% bootstrap interval of [-0.021, 0.079], so the two neutral metadata views are best treated as paired robustness views rather than separate leaderboards.

Table 5 shows that the aggregate difficulty is not uniform. Clean rows are much easier, with 0.719 operational escape. Generated-lore rows have 0.938 belief correctness but only 0.125 operational escape because evidence precision collapses to 0.000 and polluted support remains high.

Table 6 reports the same condition-level operational rates with latent-task bootstrap intervals. Generated lore remains the clearest stress case: the belief-to-operational gap is 0.812 with a 95% interval of [0.714, 0.906].

Table 5: Aggregate model metrics by evidence condition across all four models and both views.

Condition	n	Operational	Belief	Evidence precision	Polluted support
Clean	96	0.719	0.927	0.994	0.000
Buried primary	96	0.490	0.979	1.000	0.000
False consensus	96	0.302	0.719	0.659	0.323
Conflicting evidence	96	0.250	0.812	0.479	0.448
Generated lore	96	0.125	0.938	0.000	0.802

Table 6: Condition-level operational epistemic escape with 95% bootstrap intervals from latent-task resampling. Values aggregate across models and both neutral metadata views.

Condition	<i>n</i>	Operational escape
<code>clean</code>	96	0.719 [0.536, 0.886]
<code>conflicting_evidence</code>	96	0.250 [0.083, 0.433]
<code>false_consensus</code>	96	0.302 [0.135, 0.486]
<code>buried_primary</code>	96	0.490 [0.225, 0.750]
<code>generated_lore</code>	96	0.125 [0.056, 0.200]

Table 7 makes the separation explicit. The largest belief-to-operation gap appears in generated lore. Buried-primary rows also separate: belief correctness is 0.979, while operational escape is 0.490. This supports the core claim that the task is not just asking whether the model can answer the claim. It asks whether the model can keep the evidence contract clean enough for an agent pipeline.

Table 8 summarizes active-verification action behavior. Exact target rate is 1.000 in this pilot, so the remaining variation comes from required action recall, the action score, and whether the active-verification row also satisfies belief and support hygiene. This result should be read cautiously because the active-verification slice has only 96 rows.

Table 9 reports a paired Phase 1.3 robustness mini-suite over 20 frozen source tasks, two frontier models, and five presentation variants. All 200 calls parsed, and all 40 task-model groups contained the `baseline_original` row plus every perturbation row. Operational deltas were non-negative for evidence-order randomization, source-type masking, prompt paraphrase, and citation masking. The generated-lore belief/operation gap also persisted: the baseline gap was 1.000, source-type masking and citation masking remained at 1.000, and order randomization and prompt paraphrase remained at 0.900. Citation masking should be read as a stress test because visible dependency structure can be task-relevant evidence rather than neutral presentation metadata.

8 Schema/Interface Analysis

The current artifact uses the clarified schema for all main results. The earlier cued construction motivated a schema-sensitivity question, but its model results are quarantined and are not used here. After the main role-uncued pilot, we ran a small Phase 1.1 ablation over generated-lore and buried-primary rows. The ablation holds the evidence packet fixed and varies only the structured output interface. It is exploratory follow-up evidence, not a replacement for the Phase 1 main table.

The interface finding that remains valid in this pilot is narrower. Structured fields matter because evidence and action fields can fail while the final verdict succeeds. In generated-lore rows, models often know the right verdict yet fail operational escape because polluted evidence remains in support fields. In active-verification rows, the action schema makes executable next

Table 7: Belief correctness versus operational escape by condition.

Condition	Belief	Operational	Gap	Evidence precision
Generated lore	0.938	0.125	0.812	0.000
Buried primary	0.979	0.490	0.490	1.000
False consensus	0.719	0.302	0.417	0.659
Conflicting evidence	0.812	0.250	0.562	0.479
Clean	0.927	0.719	0.208	0.994

Table 8: Active-verification action exactness by model. Each model contributes 24 active-verification rows. All columns are computed on the active-verification subset.

Model	Action score	Exact target	Required recall	Active op. escape
claude-opus-4-7	0.875	1.000	0.875	0.292
gemini-3.1-pro-preview	0.812	1.000	0.812	0.417
gpt-5.5	0.792	1.000	0.792	0.542
deepseek-v4-pro	0.688	1.000	0.688	0.167

steps measurable instead of leaving them as prose.

The Phase 1.1 slice gives a concrete interface-sensitivity warning. With the minimal schema, both evaluated models placed polluted evidence in support fields on all rows in the slice and reached zero role escape. The current and clarified interfaces avoided polluted support in this slice, while the diagnostic-without-hygiene variant produced mixed role-allocation behavior. This should be read narrowly: schema wording is an experimental factor for machine-readable evidence roles. It does not show that schema wording explains all Phase 1 gaps, and it does not use the quarantined cued results.

9 Human Scorer Audit

Table 10 reports the local scorer-contract audit. The audit reviewed 40 rows balanced across models, views, conditions, task families, and operational successes/failures. It found no scorer disagreements, no systematic scorer bug, and no need for rescoring.

The audit is still limited. It is a local Codex-assisted manual scorer audit, not an independent human-subject review. The claim is therefore restricted to scorer-contract sanity for this pilot, not external validation of every action-quality judgment.

10 Limitations

The pilot is small: 60 latent tasks, two neutral metadata views, four model labels, and one prompt/schema condition. Bootstrap intervals are reported at the latent-task level, but they do not make the pilot statistically powered for fine-grained provider ranking. Reported model differences should be treated as descriptive.

The tasks are synthetic. Synthetic control is useful for isolating source-role failures, but it introduces prose regularities that real evidence environments do not share. The leakage, baseline, and template-diversity audits reduce this risk; they do not eliminate it.

The human review evidence has two different scopes. The 400-row pre-model surface review is a local leakage gate. The 40-row scorer audit is a local Codex-assisted manual audit. Neither

Table 9: Phase 1.3 paired robustness deltas relative to the within-suite baseline. Positive operational deltas mean the perturbation increased operational epistemic escape relative to **baseline_original**; positive polluted-support deltas mean more polluted support.

Variant	Pairs	Δ operational	Δ belief	Δ evidence precision	Δ polluted support
<code>order_randomized</code>	40	0.125	0.075	0.013	-0.075
<code>source_type_masked</code>	40	0.100	0.050	0.026	-0.025
<code>prompt_paraphrase</code>	40	0.100	0.025	0.025	-0.025
<code>citation_masked</code>	40	0.075	-0.075	-0.047	0.075

Table 10: Local scorer-contract audit summary.

Audit item	Value
Reviewed rows	40
Models covered	4, 10 rows each
Views covered	20 hidden, 20 visible
Conditions covered	8 rows each
Families covered	16 packet, 16 evidence, 8 active
Operational successes/failures	20 / 20
Scorer disagreements	0
Systematic scorer bug found	false
Rescoring required	false
Independent human review	false

should be described as external validation.

The Phase 1.1 schema-ablation slice is small: 16 source tasks, one visible metadata view, two main models, and a local Codex-assisted scorer audit. It supports only an interface-sensitivity claim about machine-readable evidence-role allocation. It should not be treated as a new benchmark result, a provider ranking, or evidence that schema wording explains every Phase 1 failure.

The Phase 1.3 robustness mini-suite is also narrow: 20 source tasks, one visible metadata view, two models, and controlled presentation perturbations. It supports only named mini-suite robustness or sensitivity claims. In particular, citation masking can remove task-relevant dependency information, so sensitivity under that perturbation should not be interpreted as a benchmark failure by itself. The flip review is local and scorer-assisted rather than an independent human audit.

Finally, the artifact includes hidden labels and gold data for reproducibility, but those files are not model-visible. Any downstream reuse must preserve that separation.

11 Ethics And Scope

Epistemic Hygiene Arena-Uncued is a diagnostic benchmark for controlled polluted evidence environments. It is not a safety certification, a deployment approval test, or a general leaderboard. High performance on this pilot would not establish that an agent is reliable in real legal, medical, scientific, or financial workflows.

The benchmark uses synthetic entities and synthetic evidence packets. This avoids collecting sensitive real-user data and makes gold provenance explicit. The tradeoff is ecological validity: synthetic documents cannot cover the full distribution of open-web evidence, institutional records, or adversarial manipulation.

The intended use is research analysis of failure modes: belief formation, evidence-role hygiene, shortcut sensitivity, and verification-action expression. The intended non-use is automated certifi-

cation of deployed systems without additional task coverage, independent audits, domain-specific validation, and human oversight.

12 Artifact

The release-facing artifact is rooted at `artifact_uncued_phase1/`. It includes the role-uncued pilot data, prompt artifacts, schemas, scorer files, model outputs, baselines, audits, examples, a minimal reproduction script, and an artifact verifier script. The manifest records 1022 hashed files and explicitly marks `cued_artifacts_included=false`.

The main verifier is:

```
uvruneha-verify-uncued-phase1
--artifact-dir../artifact_uncued_phase1
--reports-dir../reports
--out-dir../reports
```

The readiness verifier passes all artifact gates: layout, manifest hashes, visible-file hidden-label scan, gold/scorer inclusion, prompt inclusion, leakage gate, baseline gate, local leakage review gate, model-run completeness, scored-result completeness, scorer audit, and explicit status that the main artifact path defers schema ablation. The minimal no-dependency smoke check is `artifact_uncued_phase1/reproduce_minimal.sh`. The schema-ablation follow-up outputs are reported separately under `reports/` and `eha-mvp/results/`; they are not folded into the main pilot artifact.

The older `artifact/` package is quarantined as `cued/internal` evidence and is not a source for the main results in this paper.

13 Conclusion

Epistemic Hygiene Arena-Uncued shows that, in a controlled role-uncued pilot, answer accuracy can substantially overstate agent readiness in polluted evidence environments. The clearest evidence is generated lore: models usually reach the right belief, yet operational escape remains low because evidence roles are not clean. Active-verification rows show a second interface requirement: a useful agent must express the next verification action in a machine-checkable form.

The release should be read as a validity-first pilot. It fixes the direct cueing failure exposed by the older run, packages the uncued artifacts, and records leakage, baseline, model-run, scorer-audit, readiness, and small schema-ablation diagnostics. The next step is not leaderboard expansion alone; it is a larger role-uncued task set with independent human audit and schema/interface choices treated as experimental factors rather than neutral logging details.

A Operational Escape Scoring Specification

For a scored row r , let P_r denote parse success, B_r belief correctness, S_r clean support, and U_r uncertainty discipline. $B_r = 1$ when the predicted verdict equals the gold verdict. $S_r = 1$ when no document in `supporting_evidence` is in the task pollutant set, including contaminant, generated, or stale documents. $U_r = 1$ when the prediction is `insufficient` exactly for gold-insufficient rows and is not `insufficient` otherwise.

The diagnostic support quantities are:

$$\text{evidence precision} = \frac{|\text{supporting evidence} \cap \text{clean support}|}{|\text{supporting evidence}|},$$

when support is non-empty, and

$$\text{clean support recall} = \frac{|\text{supporting evidence} \cap \text{clean support}|}{|\text{gold clean support}|},$$

when gold clean support exists. Pollutant handling is reported through polluted support, rejected pollutant, and generated-lore avoidance metrics; it is not an unreported hidden term.

For packet-judgment rows:

$$O_r = P_r \wedge B_r \wedge S_r \wedge U_r.$$

For evidence-selection rows:

$$O_r = P_r \wedge (\text{evidence value score} \geq 0.50) \wedge (\text{duplicate avoidance} = 1) \wedge (\text{generated lore selected} = 0).$$

For active-verification rows:

$$O_r = P_r \wedge B_r \wedge S_r \wedge U_r \wedge A_r,$$

where A_r requires the relevant primary-opening, contradiction/comparison, and generated-lore trace action types for that task. Exact target rate is reported separately by matching action targets or selected documents to the required document target set.

Pseudocode.

```

pollutants = contaminant_doc_ids + generated_doc_ids + stale_doc_ids
support_dirty = any(doc in pollutants for doc in supporting_evidence)
belief_correct = predicted_verdict == gold_verdict
uncertainty_ok = (
    predicted_verdict == "insufficient"
    if gold_verdict == "insufficient"
    else predicted_verdict != "insufficient"
)

if family == "evidence_selection":
    escape = (
        evidence_value_score >= 0.50
        and duplicate_avoidance_rate == 1.0
        and not generated_lore_selected
    )
elif family == "active_verification":
    escape = belief_correct and not support_dirty and uncertainty_ok and action_ok
else:
    escape = belief_correct and not support_dirty and uncertainty_ok

operational_epistemic_escape = parse_success and escape

```

The worked examples in Appendix C show one successful clean row, one generated-lore row with correct belief but failed support hygiene, and one active-verification row with required action recall.

B Leakage And Shortcut Method Details

The automatic leakage audit scans only model-visible surfaces: task identifiers, questions, document identifiers, titles, source types, timestamps, bodies, and visible citations. It fails the gate for hidden scorer fields, direct answer cues, semantic identifiers that expose construction roles, or forbidden material-role phrases. The pilot audit reports zero critical, high, medium, hidden-label, semantic-ID, and direct-answer-cue hits.

The local surface review sampled 400 document rows covering 50 unique latent tasks. It found zero critical leaks, zero severe leakage rows, and zero role-guess hits. This was a pre-model leakage gate, not an external validation study.

The no-API shortcut baselines are: ID-only, document-order-only, title-only, source-type-only, metadata-only, timestamp-only, citation-graph-only, claim-overlap-only, cue-phrase-only, random-valid-schema, always-insufficient, and simple heuristic. The strongest shortcut is the simple heuristic, at 0.350 operational escape in the hidden view and 0.333 in the visible view. This means the benchmark is not explained by trivial labels alone, but shallow overlap can still solve enough rows that model comparisons should be phrased carefully.

The relevant reproducibility paths are `eha-mvp/eha/uncued_leakage.py`, `eha-mvp/eha/uncued_baselines.py`, `reports/eha_uncued_leakage_pilot.json`, `reports/eha_uncued_baselines_pilot.json`, and the packaged copies under `artifact_uncued_phase1/`.

C Task And Scoring Examples

The examples below expose scorer-only roles for auditability. These roles were not visible to models; model-visible packets contained only the neutral document fields.

C.1 Clean Or Buried-Primary Success

Task `uncued_000_visible`; family `packet_judgment`, condition `clean`, view `neutral_metadata_visible`.

Question Did Novalis Robotics log 184 completed checks for Atlas valve checks in March 2025?

Model `gemini-3.1-pro-preview`. Gold verdict `supported`; predicted verdict `supported`.

Output fields Support `uncued_000_d03`; selected `uncued_000_d03`; actions [{"action": "search_primary", "target": "monthly ledger for Northline service bay", "rationale": "To verify the 184 completed checks count against the primary source ledger."}].

Scorer decision Belief 1.000; evidence precision 1.000; polluted support 0.000; required action recall n/a; operational escape 1.000.

Answer excerpt Yes, based on the provided evidence, the operations timeline note confirms that the monthly ledger for the Northline service bay lists 184 completed checks for Atlas valve checks in March 2025.

uncued_000_d01 (Service Ledger Extract; timeline note; 2025-01-16; role=`context`, supports=False): A routing note lists Northline service bay as the site for the Atlas valve checks workstream. The note does not provide a closing count. The note uses the same site code, workstream name, and monthly close...

uncued_000_d03 (Operations Timeline Note; field note; 2025-01-13; role=`clean_anchor`, supports=True): The monthly ledger for Northline service bay lists 184 completed checks for

Atlas valve checks in March 2025. The entry is dated after the close of the month and uses the same work order code as the...

uncued_000_d04 (Program Registry Entry; operations memo; 2025-01-15; role=context, supports=False): A shift memo mentions that Atlas valve checks work was concentrated in the final week of March 2025. It tracks handoff timing but not the final tally. The note uses the same site code, workstream name, and...

uncued_000_d02 (Field Summary Memo; registry entry; 2025-01-14; role=context, supports=False): A planning note for Atlas valve checks describes the inspection route at Northline service bay during March 2025. It gives staffing context and does not include a final monthly count. The note uses the same...

C.2 Correct Belief With Failed Evidence Hygiene

Task `uncued_048_visible`; family `packet_judgment`, condition `generated_lore`, view `neutral_metadata_visible`.

Question Did Novalis Robotics launch the HelioSplice partner repair program in March 2025?

Model `claude-opus-4-7`. Gold verdict `insufficient`; predicted verdict `insufficient`.

Output fields Support `uncued_048_d01`, `uncued_048_d04`; selected `uncued_048_d04`, `uncued_048_d01`, `uncued_048_d03`; actions [{`"action": "trace_source"`, `"target": "uncued_048_d01"`, `"rationale": "The roundup is the originating claim; uncued_048_d04 merely cites it. Tracing to a primary operational record is needed to verify the launch."`}, {`"action": "search_primary"`, `"target": "...."`}]

Scorer decision Belief 1.000; evidence precision 0.000; polluted support 1.000; required action recall n/a; operational escape 0.000.

Answer excerpt Evidence is insufficient to confirm the launch. The only affirmative mentions (`uncued_048_d01` roundup and `uncued_048_d04` recap citing it) form a single unsourced chain, while the program registry (`uncued_048_d02`) and field memo...

uncued_048_d04 (Service Ledger Extract; registry entry; 2025-04-14; role=same_root_echo, supports=False, contamination=unsupported): A partner recap repeats the HelioSplice partner repair program launch wording from the roundup. Its visible citation points to the roundup item. The note uses the same site code, workstream name, and monthly...

uncued_048_d01 (Operations Timeline Note; field note; 2025-04-13; role=unsupported_story, supports=False, contamination=unsupported): A community roundup says the HelioSplice partner repair program appeared during March 2025. The item gives a program name and month but no linked operational record. The note uses the same site code,...

uncued_048_d02 (Program Registry Entry; timeline note; 2025-04-16; role=context, supports=False): A monthly index names the Atlas valve checks workstream and site. It lists regular service categories but no program launch entry. The note uses the same site code, workstream name, and monthly close date as...

uncued_048_d03 (Field Summary Memo; operations memo; 2025-04-15; role=context, supports=False): A service planning memo for Northline service bay lists routine Atlas valve checks work in March 2025. It does not mention the HelioSplice partner repair program. The note uses the same site code, workstream...

C.3 Active-Verification Action Case

Task `uncued_010_visible`; family `active_verification`, condition `clean`, view `neutral_metadata_visible`.

Question Did Novalis Robotics log 118 completed refurbishments for Nova pump refurbishments in July 2025?

Model `claude-opus-4-7`. Gold verdict **supported**; predicted verdict **supported**.

Output fields Support `uncued_010_d01`; selected `uncued_010_d01`, `uncued_010_d02`, `uncued_010_d03`; actions [{"action": "trace_source", "target": "uncued_010_d01", "rationale": "Verify the monthly ledger entry's provenance since it is the only document stating the 118 count."}, {"action": "search_primary", "target": "Harbor Annex July 2025 Nova pump refurbishment...."}]

Scorer decision Belief 1.000; evidence precision 1.000; polluted support 0.000; required action recall 1.0; operational escape 1.000.

Answer excerpt Yes—`uncued_010_d01` (Field Summary Memo) records 118 completed Nova pump refurbishments at Harbor Annex for July 2025, and no document contradicts this figure. Other documents corroborate the workstream and site context but lack counts....

uncued_010_d04 (Service Ledger Extract; timeline note; 2025-02-16; role=`context`, supports=False): A routing note lists Harbor Annex as the site for the Nova pump refurbishments workstream. The note does not provide a closing count. The note uses the same site code, workstream name, and monthly close date...

uncued_010_d03 (Operations Timeline Note; operations memo; 2025-02-15; role=`context`, supports=False): A shift memo mentions that Nova pump refurbishments work was concentrated in the final week of July 2025. It tracks handoff timing but not the final tally. The note uses the same site code, workstream name,...

uncued_010_d02 (Program Registry Entry; registry entry; 2025-02-14; role=`context`, supports=False): A planning note for Nova pump refurbishments describes the inspection route at Harbor Annex during July 2025. It gives staffing context and does not include a final monthly count. The note uses the same site...

uncued_010_d01 (Field Summary Memo; field note; 2025-02-13; role=`clean_anchor`, supports=True): The monthly ledger for Harbor Annex lists 118 completed refurbishments for Nova pump refurbishments in July 2025. The entry is dated after the close of the month and uses the same work order code as the...

Required action types: `open_best_visible_record`. Required target documents by view: `neutral_metadata_hidden`: `uncued_010_d03`; `neutral_metadata_visible`: `uncued_010_d01`.

D Model Invocation Details

The run recorded 480 complete model rows and no parse failures. The cost report records USD 3.764007 spent against a USD 10 soft cap, USD 15 hard cap, and USD 20 abort cap.

Table 11: Pilot model invocation profiles. Run date: 2026-05-22. All rows used the clarified schema, the standard-answer prompt, a maximum of two attempts, no LLM repair, first-JSON-object extraction, and developer/system content merged into the user message.

Model	Provider	Response format	Temperature	Max output	Parser / retry policy
<code>gpt-5.5</code>	OpenAI	<code>json_schema</code>	omitted	4096	first JSON object; max 2 attempts
<code>claude-opus-4-7</code>	Anthropic	<code>json_schema</code>	omitted	4096	first JSON object; max 2 attempts
<code>gemini-3.1-pro-preview</code>	Google	<code>json_schema</code>	omitted	4096	first JSON object; max 2 attempts
<code>deepseek-v4-pro</code>	DeepSeek	<code>json_object</code>	omitted	4096	first JSON object; max 2 attempts

E Synthetic Template Diversity Audit

The pilot balances its five conditions at 12 latent tasks each and its task families at 24 packet-judgment, 24 evidence-selection, and 12 active-verification tasks. Each neutral metadata view contains 240 documents. Body length ranges from 45 to 67 words, with median 55 and mean 54.07 words.

In the visible-metadata view, source types are balanced: 60 field notes, 60 operations memos, 60 registry entries, and 60 timeline notes. The hidden-metadata view masks source type to `document` for all 240 documents. Each view has four title strings, 36 timestamps, and 60 documents with visible citations.

The largest prose regularity is intentional: all 240 documents per view contain the neutral relation-control sentence about using the same site code, workstream name, and monthly close date. This prevents relation checks from depending on labels, but it also creates repeated 5-grams across the corpus. We therefore treat the synthetic pilot as a controlled diagnostic slice, not as evidence that template shortcuts are exhausted.

F Phase 1.3 Robustness Mini-Suite

The Phase 1.3 robustness mini-suite froze 20 role-uncued source tasks from `eha-mvp/data/uncued-pilot-v1`: five each from `generated_lore`, `buried_primary`, `conflicting_evidence`, and `false_consensus`. Within each condition, the frozen slice used two `packet_judgment` tasks, two `evidence_selection` tasks, and one `active_verification` task where available. The suite used only the `neutral_metadata_visible` view and seed 20260525.

The run compared `baseline_original` against four paired perturbations: `order_randomized`, `source_type_masked`, `prompt_paraphrase`, and `citation_masked`. The schema and scorer were held fixed to the Phase 1 interface. Pre-model gates checked that the selection manifest referenced only the role-uncued pilot data, that prompt parity changed only the intended field for each perturbation, that hidden labels and condition or family labels were absent from model-visible prompts, that projected cost stayed below the hard cap, and that every perturbation had a paired baseline row.

The full run completed 200 planned calls: 20 source tasks times two models times five variants. All 200 latest rows parsed successfully. The cost report recorded USD 2.614767 spent against a USD 7 hard cap. The prompt audit and stored hidden-label audit both passed with zero hits. The paired analysis produced 160 baseline-versus-perturbation comparisons and 57 flip-review rows for local inspection.

The run directory contains `robustness_report.md`, the paired CSV tables, model-call artifacts, prompt audits, and the flip-review sheet. The top-level reports directory contains the Markdown result summary and the machine-readable JSON summary.

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